Index Calculation Debug: `m007` (Trefoil complement)

Pipeline verification

Manifold: `m007` in the SnapPy census (figure-eight knot complement / 4_1).
Tetrahedra: $n = 3$. **Cusps:** $r = 1$.
Basis decomposition: 1 hard edge, 1 easy edge ($n - r = 2$ internal edges total).
Fugacity variables: η_1 (one per hard edge).

1 SnapPy Gluing Equations

The standard gluing equations are encoded in two integer matrices. Columns are ordered $(Z_1, Z'_1, Z''_1,\ Z_2, Z'_2, Z''_2,\ Z_3$

Internal edge equations C (3×9):

$$C = \begin{pmatrix} 1 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 2 & 0 & 2 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Cusp equations C_{cusp} $(2 \times 9$, rows μ, $\lambda)$:

$$C_{\text{cusp}} = \begin{pmatrix} 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 1 \\ 2 & 0 & 1 & 0 & -1 & -1 & -1 & 0 & -1 \end{pmatrix}$$

Row 1 is the meridian μ; row 2 is the longitude λ.

2 Geometric Basis (Hard / Easy Edges)

We choose a basis of $n - r = 2$ internal edge equations. Each row vector is a row of C (same 9-column format).

Easy edge (all non-zero entries lie in the Z_i columns, i.e. all Z'-/Z''-entries are zero):

$$v_{\text{easy}} = (1,\ 0,\ 0,\quad 2,\ 0,\ 0,\quad 1,\ 0,\ 0)$$

Hard edge (has non-zero Z'-/Z''-entries):

$$v_{\text{hard}} = (0,\ 2,\ 1,\quad 0,\ 1,\ 2,\quad 0,\ 2,\ 1)$$

Because there is one hard edge, the refined index depends on *one* fugacity variable η_1, while the easy edge contributes only to the q-power.

3 Neumann–Zagier Matrix g_{NZ} and Affine Shift

After the symmetry-reduction step the $2n \times 2n = 6 \times 6$ matrix g_{NZ} is assembled in *block form*

$$g_{\mathrm{NZ}} \;=\; \big(\; A \;\big|\; B \;\big), \qquad \text{columns: } (Z_1, Z_2, Z_3 \mid Z_1'', Z_2'', Z_3'').$$

Rows are ordered: meridian μ; hard-edge charge Q_{h}; easy-edge charge Q_{e}; longitude/2 $\lambda/2$; internal momentum Γ_1 (hard); internal momentum Γ_2 (easy).

$$g_{\mathrm{NZ}} \;=\; \left(\begin{array}{ccc|ccc} -1 & -1 & 0 & -1 & 0 & 1 \\ -2 & -1 & -2 & -1 & 1 & -1 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ \hline 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & -1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 0 & \frac{1}{2} & 0 & \frac{1}{2} \end{array}\right)$$

Affine shift. The logarithmic branch choices yield integer vectors $\boldsymbol{\nu}_X, \boldsymbol{\nu}_P \in \frac{1}{2}\mathbb{Z}^n$:

$$\boldsymbol{\nu}_X \;=\; (1,\; 3,\; -2), \qquad \boldsymbol{\nu}_P \;=\; \left(-\tfrac{1}{2},\; 0,\; 0\right).$$

Key feature: $(\boldsymbol{\nu}_P)_1 \;=\; -\frac{1}{2} \notin \mathbb{Z}$. This half-integer shift is responsible for the phase selection rule that forces all odd-m contributions to vanish in the unrefined (e=0) index. Note: when the local charges $\widetilde{m}_k, \widetilde{e}_k$ are all integers, the phase exponent φ is *automatically* an integer (a structural consequence of the NZ matrix). The *only* integrality constraint on the summation is that $g_{\mathrm{NZ}}^{-1}\kappa \in \mathbb{Z}^{2n}$.

4 Inverse Matrix g_{NZ}^{-1} and Local Charges

4.1 The inverse matrix

$$g_{\mathrm{NZ}}^{-1} \;=\; \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} & -1 & 1 & 0 \\ -1 & 0 & -\frac{1}{2} & -1 & -2 & 1 \\ -\frac{1}{2} & 1 & \frac{3}{2} & -1 & -1 & 2 \\ \frac{1}{2} & 0 & 0 & 0 & -2 & 1 \end{pmatrix}$$

4.2 Master charge vector

The charge vector entering the summation is

$$\kappa \;=\; \begin{pmatrix} m \\ 0 \\ 0 \\ e \\ e_1 \\ e_2 \end{pmatrix},$$

where $m \in \mathbb{Z}$ is the meridian charge, $e \in \mathbb{Z}$ is (half) the longitude charge, the two zeros fix the position charges of the internal edges (always zero in the summation), and $e_1, e_2 \in \frac{1}{2}\mathbb{Z}$ are the momenta of the hard and easy internal edges, respectively. The hard-edge momentum e_1 contributes the fugacity $\eta_1^{2e_1}$ to every term.

4.3 Local tetrahedron charges

Set $\Delta = g_{\mathrm{NZ}}^{-1}\,\kappa$. The local charges split as $(\widetilde{m}_1, \widetilde{m}_2, \widetilde{m}_3,\, \widetilde{e}_1, \widetilde{e}_2, \widetilde{e}_3)$:

$$\begin{aligned}
\widetilde{m}_1 &= \tfrac{m}{2} + e + e_1,\\
\widetilde{m}_2 &= -e_1,\\
\widetilde{m}_3 &= -\tfrac{m}{2} - e + e_1,
\end{aligned}$$

$$\begin{aligned}
\widetilde{e}_1 &= -m - e - 2e_1 + e_2,\\
\widetilde{e}_2 &= -\tfrac{m}{2} - e - e_1 + 2e_2,\\
\widetilde{e}_3 &= \tfrac{m}{2} - 2e_1 + e_2.
\end{aligned}$$

4.4 Phase exponent

Each term in the index carries the factor $(-1)^{\varphi}\, q^{\varphi/2}$ from the affine shift:

$$\begin{aligned}
\varphi &= \sum_{k=1}^{3} \widetilde{m}_k\, (\boldsymbol{\nu}_P)_k - \sum_{k=1}^{3} \widetilde{e}_k\, (\boldsymbol{\nu}_X)_k\\
&= \widetilde{m}_1 \cdot \left(-\tfrac{1}{2}\right) - \widetilde{e}_1 \cdot 1 - \widetilde{e}_2 \cdot 3 - \widetilde{e}_3 \cdot (-2).
\end{aligned}$$

Substituting the local charges gives the closed form

$$\varphi \;=\; \frac{13m}{4} + \frac{7e}{2} + \frac{e_1}{2} - 5e_2.$$

The *only* constraint on the summation is that the local charges $\widetilde{m}_k, \widetilde{e}_k \in \mathbb{Z}$ (i.e. $g_{\mathrm{NZ}}^{-1}\kappa \in \mathbb{Z}^{2n}$). Whenever this holds, $\varphi \in \mathbb{Z}$ automatically, so no separate $\varphi \in \mathbb{Z}$ filter is needed or applied.

5 Refined Index Formula and Truncated Output

5.1 Index formula

$$\mathcal{I}^{\mathrm{ref}}(m, e; \eta_1; q) \;=\; \sum_{\substack{e_1,\, e_2 \,\in\, \frac{1}{2}\mathbb{Z}\\ g_{\mathrm{NZ}}^{-1}\kappa \,\in\, \mathbb{Z}^{2n}}} \eta_1^{2e_1}\, (-1)^{\varphi}\, q^{\varphi/2} \prod_{k=1}^{3} \frac{(-1)^{\widetilde{m}_k} q^{\widetilde{m}_k/2}}{1 - (-1)^{\widetilde{m}_k + \widetilde{e}_k} q^{(\widetilde{m}_k + \widetilde{e}_k)/2}}.$$

The product of three tetrahedron factors is $\displaystyle\prod_{k=1}^{3} \mathcal{T}(\widetilde{m}_k, \widetilde{e}_k; q)$, where the single-tetrahedron building block is

$$\mathcal{T}(m', e'; q) \;=\; \frac{(-1)^{m'} q^{m'/2}}{1 - (-1)^{m'+e'} q^{(m'+e')/2}}.$$

5.2 Truncated expansion at $(m, e) = (0, 0)$

Computed to order q^6 (half-integer power cut-off 12):

$\mathcal{I}^{\mathrm{ref}}(0, 0; \eta_1; q) \;=\; 1 - q(2+\eta_1) + q^2(\eta_1^{-1} - 2 - \eta_1) + q^3(2\eta_1^{-1} + 4 + 2\eta_1 + \eta_1^2) + q^4(\eta_1^{-1} + 9 + 7\eta_1 + 2\eta_1^2) + q^5(-4\eta_1^{-1} + 13 + 12\eta_1 +$

Remark: The coefficient of q^6 at η_1^0 is *zero* (no constant-η_1 term appears at order q^6).

The full coefficient table at $(m, e) = (0, 0)$:

q-power	η_1-power	coefficient
q^0	η_1^0	1
q^1	η_1^0	-2
q^1	η_1^1	-1
q^2	η_1^{-1}	1
q^2	η_1^0	-2
q^2	η_1^1	-1
q^3	η_1^{-1}	2
q^3	η_1^0	4
q^3	η_1^1	2
q^3	η_1^2	1
q^4	η_1^{-1}	1
q^4	η_1^0	9
q^4	η_1^1	7
q^4	η_1^2	2
q^5	η_1^{-1}	-4
q^5	η_1^0	13
q^5	η_1^1	12
q^5	η_1^2	4
q^6	η_1^{-1}	-11
q^6	η_1^0	0 (absent)
q^6	η_1^1	11
q^6	η_1^2	2
q^6	η_1^3	-1